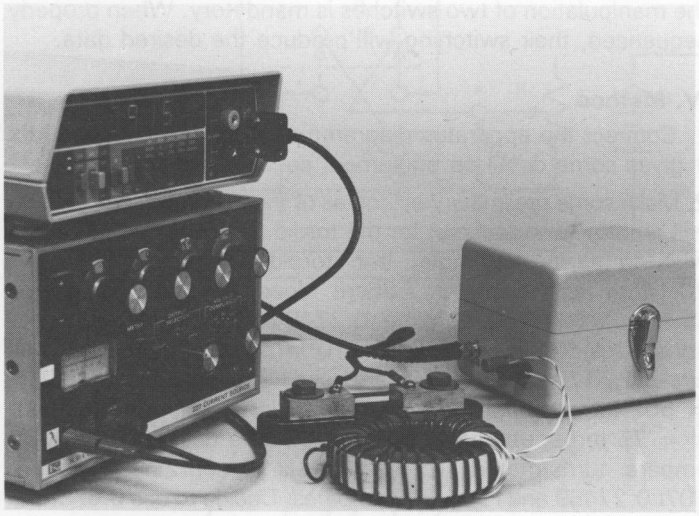
Physics Experiment #2 (Series II)
The Determination of the B vs. H Curves and the Hysteresis Loop of Magnetic Materials by the Rowland Ring Ballistic Step Method



I. Objective

Toroidal (''doughnut shaped'') cores are the most efficient form of any core configuration. Iron powder and ferrite toroids are used extensively in modern electronic equipment. They are self-shielding and useful up to the 200MHz frequency range. The measurement of the magnetic properties of materials began with Henry J. Rowland (1848-1901), American physicist, who was made famous for his contributions to spectroscopy by his invention of the modern diffraction grating and its ruling machine. This experiment has as its objective to follow faithfully the ''ballistic step method'' of Rowland, modernizing his ballistic galvanometer readout through use of an IC integrator and a Keithley DMM.

FIGURE 1. The B vs. H Curve

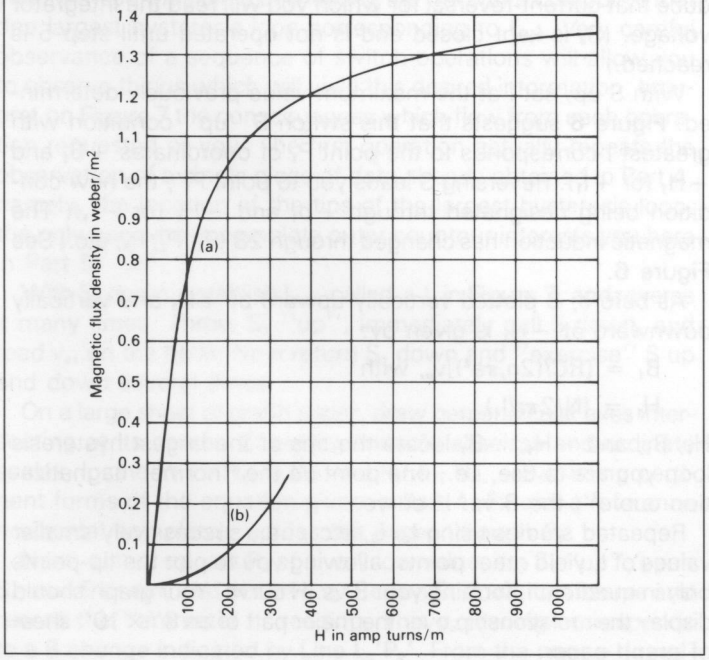
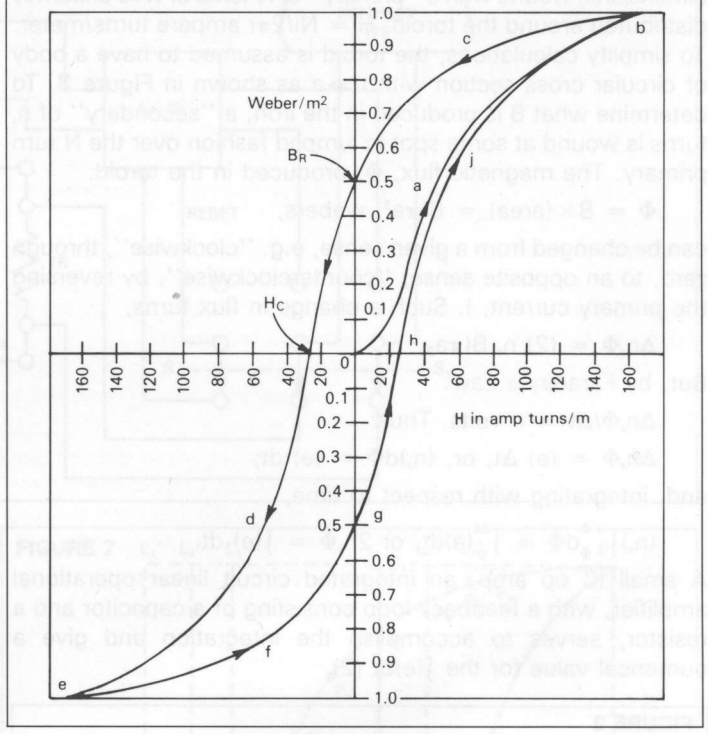


FIGURE 2. The Hysteresis Loop



II. Theory

The magnetic field vector, $\vec{B}$, also called ''magnetic induction'', is measured in ''tesla'', or equivalently in webers/m². Magnetic flux, Φ, passing through a surface, results from summing B over that surface, i.e., $\Phi = \int \vec{B} \cdot d\vec{S}$, with the unit of magnetic flux, Φ, being the ''weber''.

The quantitative relationship between current, i, and magnetic induction, $\vec{B}$, is given by Ampere's Law:

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_o i$$

for the steady state, where $\mu_o = 4\pi \times 10^{-7}$weber/(amp. meter), and the line integral is evaluated around a closed loop path through which the current, i, crosses. Since for an empty toroid wound with a total of N turns, the current crosses the circle line (the B path) N times, the line integral reduces to:

$$(B) \times (2\pi r) = N\mu_o i, \text{ or}$$

$$B/\mu_o = |H| = (Ni/2\pi r)(\text{amp. turns})/\text{meter}$$

In this last equation, by assuming an ''empty'' toroid winding and dividing B_{vacuum} by μ_o, the magnetic intensity, H, has been calculated. When a material fills the toroid interior, the same relationship, $H = Ni/2\pi r$, gives the magnetic intensity, but the corresponding magnetic induction, B, which this produces in the material core, depends on its permeability, μ, defined now through $B = \mu H$. These are convenient and important equations because, in the special case of a toroid, we can calculate H directly from the measured current in the winding, the number of turns of wire, and the measured radius of the toroid (1).

In ferromagnetic materials, μ has a very complex dependence on H. A further complication in this non-linear relationship results from a dependence on the previous magnetic history of the material being measured. Thus, for a rather complete description, both a ''B vs. H curve'' and one (or several) ''hysteresis loops'' must be experimentally determined. Typical graphs for these ''B in relation to H'' presentations are swown in **Figures 1 and 2**, which are typical of the silicon steels much used in ac power transformers.

Given an iron (solid, formed powder, or ferrite) toroid of known dimensions, wound with a "primary" of N turns of wire uniformly distributed around the toroid, $H = Ni/2\pi r$ ampere turns/meter. To simplify calculations, the toroid is assumed to have a body of circular cross section with $r \gg a$ as shown in **Figure 3**. To determine what B is produced in the iron, a "secondary" of n_s turns is wound at some spot in lumped fashion over the N turn primary. The magnetic flux, Φ, produced in the toroid:

$$\Phi = B \times (\text{area}) = B(\pi a^2) \text{ webers,}$$

can be changed from a given sense, e.g. "clockwise", through zero, to an opposite sense, "counterclockwise", by reversing the primary current, I. Such a change in flux turns,

$$\Delta n_s \Phi = (2)\, n_s\, B(\pi a_2).$$

But, by Faraday'a Law:

$\Delta n_s \Phi / \Delta t = e$ volts. Thus:

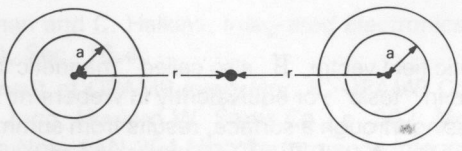
$\Delta n_s \Phi = (e)\, \Delta t$, or, $(n_s) d\Phi = (e)\, dt$;

and, integrating with respect to time,

$$(n_s) \int_{-\Phi}^{+\Phi} d\Phi = \int_0^{\Delta t} (e)dt, \text{ or } 2n_s\Phi = \int (e)\, dt.$$

A small IC op amp, an integrated circuit linear operational amplifier, with a feedback loop consisting of a capacitor and a resistor, serves to accomplish the integration and give a numerical value for the $\int (e)dt$.(2)

FIGURE 3



Consider the operational amplifier circuitry of **Figure 4a** and its equivalent representation in **Figure 4b**.

FIGURE 4a

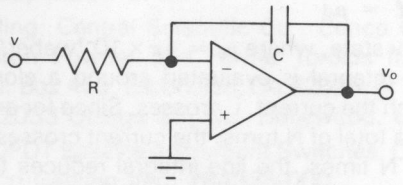
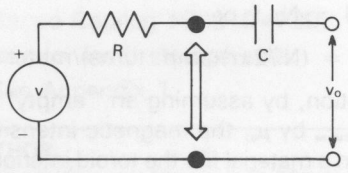


FIGURE 4b



(The double bar arrow represents a "virtual ground). Here,

$$i = v/R \text{ and } v_o = (-1/C)\int i dt = (-1/RC)\int v dt$$

The amplifier, therefor, provides an output voltage proportional to the integral of the input voltage. (3)

In the application here, v_o, the output of the integrator, would be read with a DMM, multiplied by RC, and equated to the needed $\int (v)dt$. Thus:

$$2(n_s\Phi) = \int (v)dt \text{ becomes } (2)(n_s)(B\pi a^2) = RCv_o, \text{ and}$$

$$B = [RC/(2n_s\pi a^2)]v_o \text{ telsa, with}$$

$$H = (N/2\pi r)\, I \text{ (ampere turns)/meter}$$

III. Procedure

The "magnetization curve" is a graph showing the magnetic induction, B, produced in a previously demagnetized material as the magnetic intensity, H, is increased upward from zero. (See **Figure 1**.)

A "hysteresis loop" similarly relates B values to H values when the latter are varied cyclically from a specified maximum in one sense to an equal reversed value of magnetic intensity. (See **Figure 2**.) However, for the first, the B vs. H curve, a method for deriving acceptable **Figure 1** data to produce the "normal magnetization curve" uses the "half-displacement", (½) × (2B), between the tips of successively smaller hysteresis loops. In this method, a careful following of the directions concerning the manipulation of two switches is mandatory. When properly sequenced, their switching will produce the desired data.

IV. Method

1. Connect the apparatus diagrammed in **Figure 5** (**Appendix 1** gives some detail on equipment specifications).

2. Make some preliminary estimates of the maximum current you will employ (and reverse) for the toroid sample of material you wish to study. **Example:** Iron toroid #H. Outer diameter: 10.14cm. Inner diameter: 7.36cm. Mean radius: 4.375cm (peripheral). Mean circumference: 0.27489m. Radius of cross-sectional circle: $0.695\text{cm} = 6.95 \times 10^{-3}\text{m}$. Area of cross-sectional circle: $1.5175 \times 10^{-4}\text{m}^2$. Primary winding (evenly distributed): N = 507 turns, about #20 wire. Secondary winding (bunches): N = 75 turns, about #30 wire. Assume a desired H of 1000 ampere turns/meter. Then, from $H = NI/2\pi r$, $1000 = 507I/0.27489$ and I = 0.542 amperes. Decide to use 0.6A maximum, and to vary it downward in steps of about 0.05A.

3. Directions for Securing Data for a B vs. H Curve
Closing the "reversing" switch S "up" causes current to flow, producing toroid flux Φ in one sense. Function switch S_1, when pulled down, connects the DMM across the standard 1Ω shunt, thereby converting its digital readout to amperes (1 ampere for 1 volt, etc.); when thrown up, S_1 connects the DMM to read to volts output of the integrator, the "v_o" of the equations presented. Before each "up use of flux change, the reversing switch S should be "exercised" up and down several times to establish a history of excursions stabilized around the outer confines of that one hysteresis loop being studied.

4. Produce the data for the normal magnetization curve, B vs. H.
Keep S_1 down at all times except when you are about to produce that current-reversal for which you will read the integrator voltage. (S_4 is kept closed and is not operated until step 5 is reached.)

With S up, set I at the maximum value previously determined. **Figure 6** suggests that this switch S "up" condition with greatest I corresponds to the point P_1 of coordinates $+B_1$ and $+H_1$ (or $+I_1$). Reversing S leads you to point P_1', the new condition being designated through $-b_1$ and $-H_1$ (or $-I_1$). The magnetic induction has changed through 2B (P_1, P_2, P_3, etc.) See **Figure 6**.

As before, B plotted vertically upward at $+H_1$ and vertically downward at $-H_1$ is given by:

$$B_1 = [RC/(2n_s\pi a^2)]v_o, \text{ with}$$

$$H_1 = (N/2\pi r/I_1).$$

H_1, B_1, and $-H_1$, $-B_1$ locate the tips of the largest hysteresis loop you are to use, i.e., **one** point on the "normal magnetization curve", the B vs. H curve.

Repeated studies using I_2, I_3, etc., i.e., successively smaller values of I, yield other points, allowing you to plot the tip-points only in quadrant I, forming your B vs. H curve. Your graph should display their relationship over the major part of an 8" × 10" sheet of graph paper.

FIGURE 5

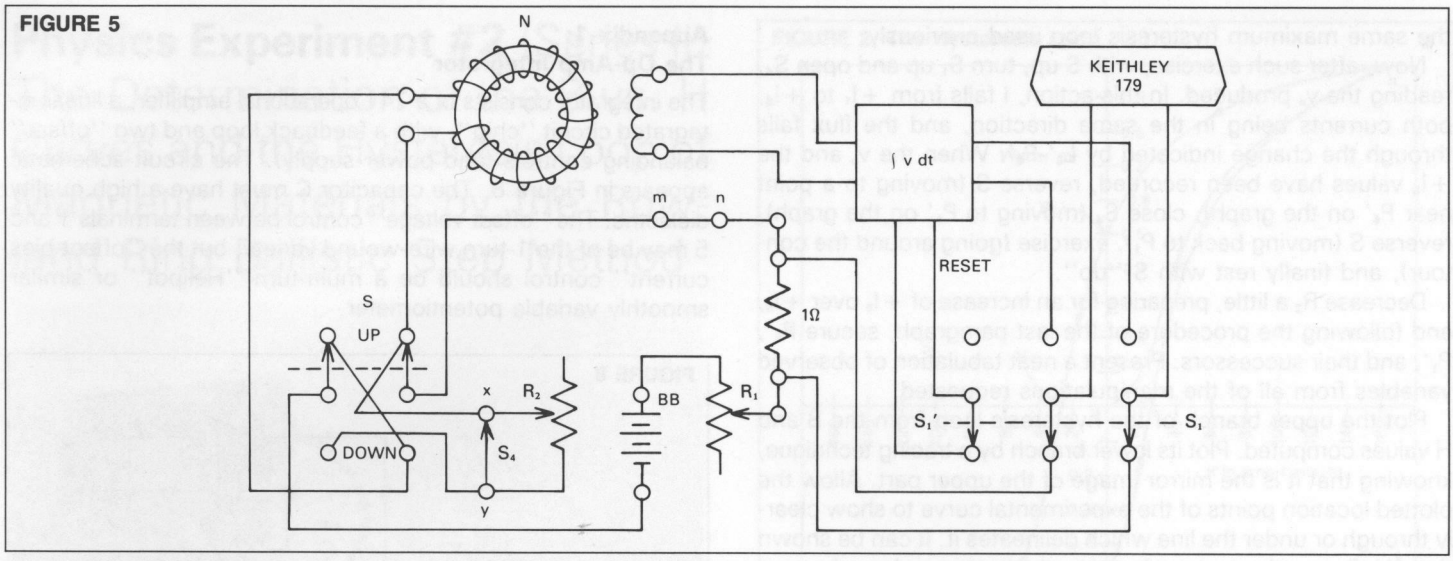


FIGURE 6

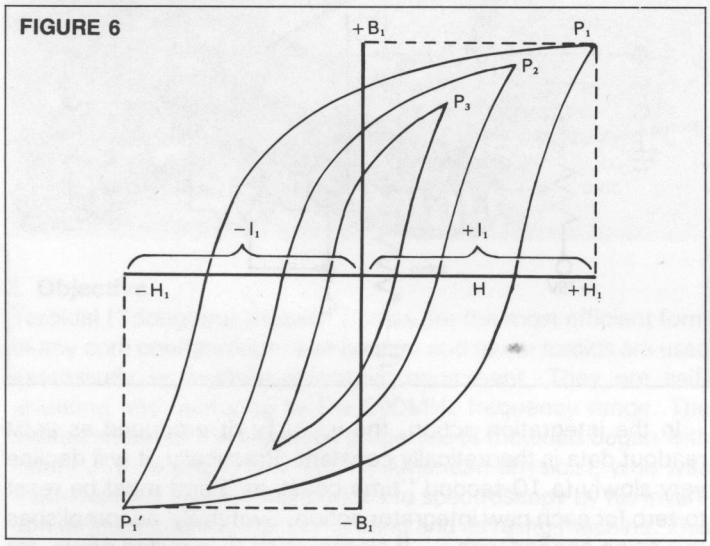
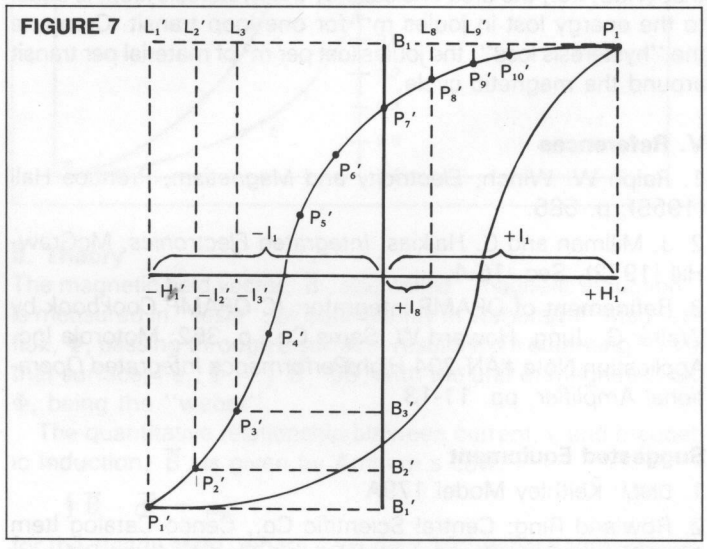


FIGURE 7



5. Produce the data for the determination of the hysteresis loop, cyclic B vs, H.

Though Part 4 gave you tips of successive hysteresis loops, you still need the points delineating the the entire contour of that one largest hysteresis loop corresponding to I_{max}. Very careful observance of a sequence of switch operations will allow you to observe things which will yield this desired information. Interpret on **Figure 7** the consequences which flow from each operation requested of you. The first operation actually repeats the observation of a single piece of data already obtained in Part 4, namely, the location of the tips of the largest hysteresis loop, the only one whose complete outer countour interests you here in Part 5.

With S_1 down, establish I_{max}, called $+I_1$ in **Figure 7**, and reverse it many times. Throw S_1 ''up'', immediately pull S down, and read v_{o1} on the DMM. Now return S_1 down and ''exercise'' S up and down several times.

On a large sheet of graph paper, draw perpendicular axes intersecting at its center. Choosing suitable abscissa and ordinate scales, locate P_1 $(+H_1, +B_1)$ and P_1' $(-H_1, -B_1)$, using pertinent forms of the equation given in Part 4. Extend a reference horizontal completely across the paper through P_1.

Next, with S up, set R_2 at a very low value, set S_1 ''up'', open S_4, pull S down, and observe v_o. Return S_1 down to observe and record the ''ammeter I'' reading. This v_o reading corresponds to a B change indicated by Line $L_2'P_2'$. From the new $-I_2$ reading (less than I_{max} because of R_2), you can compute $-H_2$, and from the v_{o2} you can compute the B change and therefore place point p_2' correctly on your graph.

S is still down. **CAUTION!** (Interpret on **Figure 5** the consequences of each step to be taken.) Close S_4, thereby going to P_1' on the graph. Push S up, thereby going back to P_1 on the graph. Now exercise through a few full cycles by repeated reversals on switch S, ending in its ''up'' position.

For a new point such as P_3' with S up, make R_2 still a little larger, open S_4, and after setting S_1 ''up'', pull S down to observe v_{o3}. Return S_1 down and read the still smaller $-I_3$ value, and follow the directions already detailed (S_4 closure before S reversal) to preserve the motion on the graph solely along the outer contour of the same, single, large hysteresis loop. Repeat the whole of the procedure, securing P_4', P_5', etc.

To secure P_7', after the cycle ''exercise'', after shifting S_1 ''up'', move S from its ''up'' to its ''open'' (mid or intermediate) position, reading v_o. The current, of course, drops to zero. Pull S_1 down and continue the motion down, then up, and exercise.

To secure P_8' and its successors, you must slightly modify the circuit, but while doing so, you should maintain condition ''P_1'', that is, maintain $+I_1$, the maximum current in the original S ''up'' sense. To do this, substitute a shorting wire at x-y in the S diagonal in place of the S_4-R_2 assembly. Connect this S_4-R_2 assembly into the current supply circuit at m-n. With S_4 closed and R_2 high, reversals of S will carry the sample around

the same maximum hysteresis loop used previously.

Now, after such exercise, with S up, turn S_1 up and open S_4, reading the v_o produced. In this action, I falls from $+I_1$ to $+I_8$, both currents being in the same direction, and the flux falls through the change indicated by L_8'-P_8'. When the v_o and the $+I_8$ values have been recorded, reverse S (moving to a point near P_6' on the graph), close S_4 (moving to P_1' on the graph), reverse S (moving back to P_1), exercise (going around the contour), and finally rest with S ''up''.

Decrease R_2 a little, preparing for an increase of $+I_9$ over $+I_8$, and following the procedure of the last paragraph, secure P_9', P_1', and their successors. Present a neat tabulation of observed variables from all of the manipulations requested.

Plot the upper branch of the hysteresis loop from the B and H values computed. Plot its lower branch by a tracing technique, knowing that it is the mirror image of the upper part. Allow the plotted location points of the experimental curve to show clearly through or under the line which delineates it. It can be shown that $\oint HdB$, i.e., the area enclosed by the hysteresis loop, is equal to the energy lost in joules m^{-3} for one loop transit. Compute the ''hysteresis loss'', the joules lost per m^3 of material per transit around the magnetic cycle.

V. References

1. Ralph W. Winch, *Electricity and Magnetism*, Prentice Hall (1955), p. 585.

2. J. Millman and C. Halkias, *Integrated Electronics*, McGraw-Hill (1972), Sec. 16.4.

3. Refinement of OPAMP integrator: *IC OPAMP Cookbook* by Walter G. Jung, Howard W. Sams Co., p. 362; Motorola Inc. Application Note #AN-204 *High Performance Integrated Operational Amplifier*, pp. 11-13.

Suggested Equipment

1. DMM: Keithley Model 179A

2. Rowland Ring: Central Scientific Co., Cenco Catalog Item #78535, or Iron Powder and Ferrite Toroids from Palomar Engineers, P.O. Box 455, Escondido, CA 92025, or from Amidon Associates, 12033 Otsego St., No. Hollywood, CA 91607.

3. Switches: DPDT Bakelite Instrument Switches, Central Scientific Co., Cenco Catalog Item #84335-000.

4. Rheostats: Cenco Catalog #82910 (12Ω.

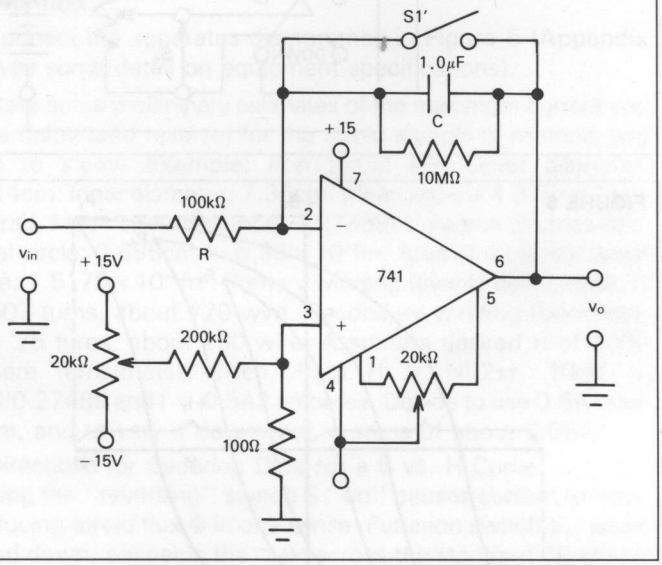
5. Battery: Edison (or lead) storage cells.

6. Integrator: See Appendix 1.

Appendix 1: The Op-Amp Integrator

The integrator consists of a 741 operational amplifier, a linear integrated circuit ''chip'', with a feedback loop and two ''offset'' balancing controls (and power supply). The circuit schematic appears in **Figure 8**. The capacitor C must have a high quality dielectric. The ''offset voltage'' control between terminals 1 and 5 may be of the 1-turn wire-wound variety, but the ''offset bias current'' control should be a multi-turn ''Helipot'' or similar smoothly variable potentiometer.

FIGURE 8



In the integration action, the $v_o = \int v\,dt$ produced as DMM readout data is theoretically constant. Practically, it will decline very slowly (a 10-second ''time constant'') and must be reset to zero for each new integrator action. Switch S_1' accomplishes this being ganged with S_1. It closes when S_1 is pulled down, for which placement current I is read by the DMM.

The ± 15 volt regulated power supply may be a miniature, like the Lambda Corp. Model LZD-12. The 10MΩ resistor across the integrating capacitor serves with the input resistor R to limit the open loop dc gain of the amplifier, assisting in the attainment of stability. At the same time, the long 10-second RC time constant affords ample time for reading v_o before its decrease due to capacitor discharge.

ABOUT THE AUTHOR

Dr. William J. Hamm, Professor of Physics at St. Mary's University, obtained his Bachelor of Science degree from the University of Dayton in 1930; his Master's Degree from the Catholic University, Washington, DC, in 1935; and his Ph.D. from Washington University, St. Louis, in 1942.

Dr. Hamm began his career of teaching college physics as an instructor at St. Mary's University in 1935. His work as an educator won for him the Minnie Stevens Piper Foundation Award as ''Piper Professor'' in 1958. He was elected a Fellow of the Institute of Radio Engineers in 1960.

In continuing studies, particularly during the summers, Dr. Hamm has pursued special courses, or done research, at a dozen different universities including DePaul, UCLA, Chicago, St. Louis, Oklahoma, Berkeley, Beloit, The Oak Ridge Institute of Nuclear Studies, Michigan State, Tufts, Colorado State, and Stanford.

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